Scarlet Ferguson

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EDUCATION

Lehigh University Bethlehem, PA
Bachelor's Degree in Computer Science and Engineering. *May 2024*
Relevant Coursework: AI Game Programming, Capstone Project (VR Game), Mobile Robotics, Programming Languages, Web Systems Programming, Fundamentals of Internetworking

SKILLS

Programming Languages: C++, C, Python, Java, JavaScript, HTML, CSS, SQL, Dart, Scala, Scheme, Markdown
Tools and Platforms: Godot, Unity, GitHub, Visual Studio Code, Android Studio, Arduino, SQL Developer, Jekyll

EXPERIENCE

Tech Consultant December 2025 – May 2026
Segura HR *Allentown, PA*
- Researched and evaluated technology and AI services to support business consulting engagements
- Compiled and summarized employment and tax compliance documentation for client use
- Compiled and summarized public health compliance documents

IT Support Specialist and HR Assistant November 2025 – January 2026
Promise Neighborhoods *Allentown, PA*
- Provided front-line technical support installing, configuring, and maintaining computers, peripherals, mobile devices, and software applications
- Maintained accurate and organized employee personnel files, both digital and physical
- Performed periodic file audits to ensure completeness and correctness

STEM Instructor / IT Intern Summer 2023
Lavner Education *Norristown, PA*
- Set up and configured computers and peripherals for camp activities
- Taught STEM curriculum to children ages 7-10

PROJECTS

Represent (WIP)
- Independently designing and developing a 2D platform shooter with RPG elements and a dystopian sci-fi setting in Godot
- Building toward a full playable demo level; additional details available on personal website

Realm of the Divine Beasts (Alpha)
- Designed an expansive alternate ruleset to Dungeons and Dragons 5th Edition, combining 5e, 3.5, and original worldbuilding
- Handbook is available on personal website and provides a fully playable tabletop role-playing game (TTRPG)

Capstone Project: VR Environmental Education Game Spring – Fall 2023
Lehigh CSE 280 and 281
- Contributed to a 2-semester, 3-person team project building a VR game in Unity to educate the public about the Lehigh watershed
- Gameplay involved surveying macroinvertebrates at multiple river locations to illustrate the effects of different types of pollution
- Implemented and documented hand-tracking and object interaction; troubleshot issues with VR hardware
- Researched project-relevant topics and compiled information to support team development decisions

AI Game Programming Projects Spring 2024
Lehigh CSE 348
- Completed individual projects involving NPC pathing, decision-making, and procedural content generation
- Implemented search algorithms including breadth-first, depth-first, and best-first search
- Implemented decision-making algorithms including behavior trees, decision trees, and Finite State Machines

Mobile Robotics Projects Spring 2024
Lehigh CSE 360
- Completed individual and team projects involving the development, programming, and testing of an airborne drone
- Programmed manual ground controls to maneuver the drone remotely through obstacles
- Implemented automatic height adjustment, visual target tracking and pursuit, and autonomous searching

Summer Phone Blues (Link to the Full Album)
- Self-produced and self-published album of original music composed from my time in high school through college, spanning classical to electronic genres
- Full album available on YouTube